



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
27-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS & TNAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

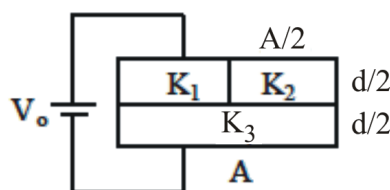
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

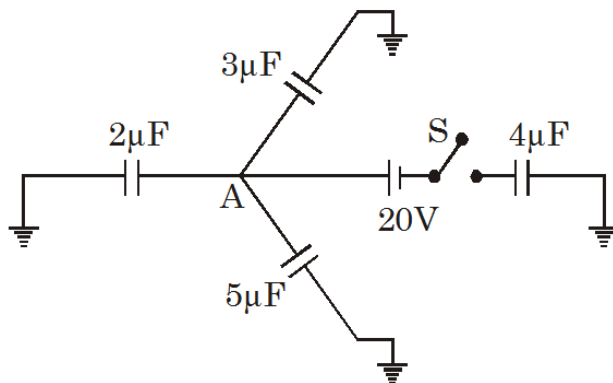
1. An ideal parallel plate capacitor of area A is filled with three dielectric slabs having dielectric constants $K_1 = 3.0$, $K_2 = 5.0$ and $K_3 = 2.0$ as shown in the figure. Choose the **CORRECT** option(s) :-



- (A) the equivalent capacitance of the system is $\frac{8\epsilon_0 A}{3d}$
- (B) the electric field in K_1 is more than the electric field in K_2 .
- (C) the electric field on the left half of K_3 is less than the electric field on the right half of K_3 .
- (D) the electric field on the left half of K_3 is more than the electric field on the right half of K_3 .

Space for Rough Work

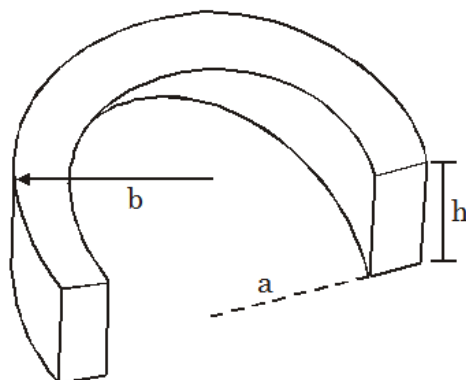
2. Three capacitors of $2\mu\text{F}$, $3\mu\text{F}$ and $5\mu\text{F}$ are individually charged with batteries of emf's 5V , 20V and 10V , respectively. After disconnecting from the voltage sources, these capacitors are connected as shown in the figure with their positive polarity plates connected to A and their negative polarity plates being earthed. Now, a battery of 20V and an uncharged capacitor of $4\mu\text{F}$ capacitance are connected to the junction A with a switch S, as shown. The switch is closed.



- (A) The potential of the junction A becomes $\frac{100}{7}\text{V}$
- (B) Final charge on the capacitor of lowest capacitance is $200\mu\text{C}$
- (C) The total charge flow to the ground after closing the switch is $200\mu\text{C}$
- (D) The charge flow to the ground after closing the switch is zero.

Space for Rough Work

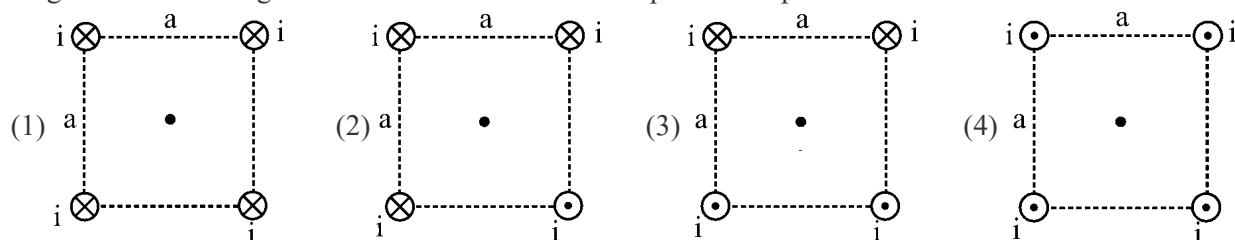
3. A conductor having resistivity ρ is bent in the shape of a half cylinder as shown in the figure. The inner and outer radii of the cylinder are a and b respectively and the height of the cylinder is h . A potential difference V is applied across the two rectangular faces of the conductor using a cell of emf V . [Neglect the internal resistance of the cell and ignore any role of the magnetic field due to the current]



- (A) Resistance offered by the conductor is $\frac{\rho \pi b}{(b-a) h}$.
- (B) Current flowing through the cell is $\frac{V (b-a) h}{\rho \pi b}$.
- (C) Current flowing through the cell is $\frac{V \cdot h \cdot \ln \left(\frac{b}{a} \right)}{\rho \cdot \pi}$
- (D) Outer cylindrical surface is at a lower voltage than the inner surface.

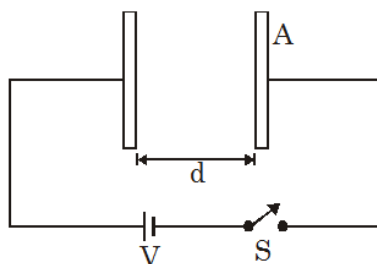
Space for Rough Work

4. Four long wires, with identical currents, either directly into or directly out of the page, are placed at four corners of a square (of side a) in four different arrangement as shown. B_1 , B_2 , B_3 and B_4 are the magnitude of net magnetic field at the center of the square in respective cases.



- (A) $B_1 = B_4 = 0$ (B) $B_2 = \frac{\sqrt{2}\mu_0 i}{\pi a}$ (C) $B_3 = \frac{2\mu_0 i}{\pi a}$ (D) $B_2 = B_3$

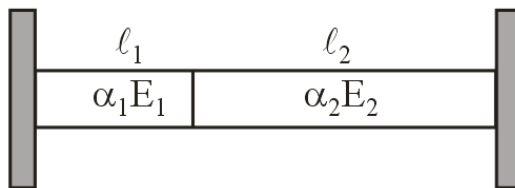
5. A parallel plate air capacitor has plate area A and separation between the plates d . Switch S is closed to connect the capacitor to a cell of emf V .



- (A) Amount of heat generated in the circuit as the capacitor gets charged is $\frac{1}{2}CV^2$
 (B) Magnitude of force that one capacitor plate exerts on the other is $\frac{\epsilon_0 AV^2}{d^2}$
 (C) If the distance between the plates is slowly reduced to $\frac{d}{2}$, the work done by external agent in the process is given by $-\frac{1}{2}CV^2$
 (D) If the distance between the plates is reduced slowly from d to $\frac{d}{2}$, energy dissipated in the circuit is zero.

Space for Rough Work

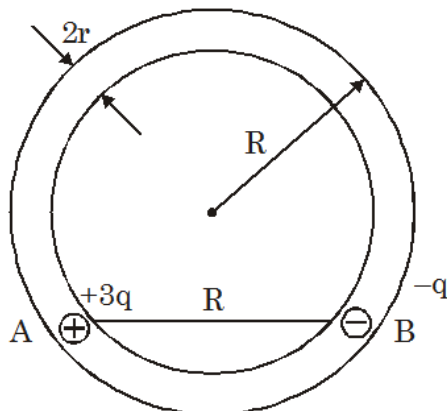
6. Two rods of different materials are placed between massive walls as shown in figure. The cross section of the rods is A their moduli of elasticity are E_1 and E_2 respectively. If rods are heated by t degrees, then (coefficients of linear expansion of material of rods are α_1 and α_2 respectively)



- (A) the strain produced in both rods is equal
- (B) the stress produced in both the rods is equal
- (C) the force exerted by one rod on the other is $\frac{\alpha_1 \ell_1 + \alpha_2 \ell_2}{\frac{\ell_1}{E_1} + \frac{\ell_2}{E_2}} A t$
- (D) the force exerted by one rod on the other is $\frac{(E_1 - E_2)^2 A}{E_1 + E_2}$
7. Two satellites A and B move round the earth in the same orbit. The mass of B is three times the mass of A.
- (A) Speeds of A and B are different
- (B) The potential energy of earth +A is same as that of earth +B.
- (C) The kinetic energy of A and B are different.
- (D) The total energy of earth +A is same as that of earth +B.

Space for Rough Work

8. On a horizontal table, there is a smooth circular groove of mean radius R . The walls of the groove are non conducting. Two metal balls A and B (each having mass m and radius r) are placed inside the groove with their centers R apart. The balls just fit inside the groove. The two balls are given charge $+3q$ and $-q$ are released from state of rest as shown in figure. Ignore the non-uniformity in charge distribution as the balls come close together and collide. The collision is elastic.



- (A) Speed of ball A is just before first collision is $\sqrt{\frac{3q^2}{8\pi\epsilon_0 m} \left(\frac{1}{2r} - \frac{1}{R} \right)}$
- (B) Speed of ball B is just before first collision is $\sqrt{\frac{3q^2}{4\pi\epsilon_0 m} \left(\frac{1}{2r} - \frac{1}{R} \right)}$
- (C) Maximum speed acquired by each ball after they collide for the first time is $\sqrt{\frac{q^2}{8\pi\epsilon_0 m r R} (4R - 7r)}$
- (D) Maximum speed acquired by each ball after they collide for the first time is $\sqrt{\frac{q^2}{4\pi\epsilon_0 m r R} (4R - 7r)}$

Space for Rough Work

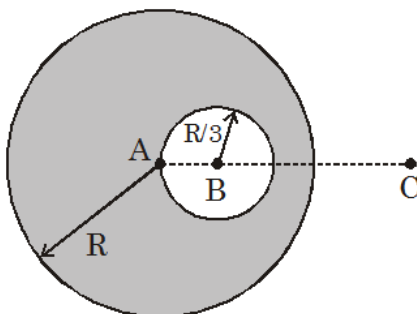
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

There is a spherical cavity of radius $R/3$ in a spherical solid conductor of radius R . A is centre of solid conductor, B is centre of cavity and C is a point outside at a distance $R/3$ from the surface. In List-I, there are certain charge configuration suggested and List-II discusses the electric field and potential.



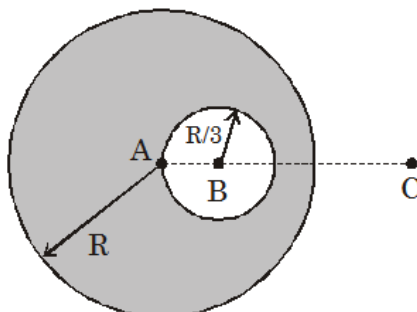
List-I		List-II	
(I)	Charge +Q at B	(P)	Electric field in cavity near A is to left
(II)	Charge +Q at C	(Q)	Electric field in cavity near A is to right
(III)	Charge -Q at B	(R)	Electric potential at A is +ve
(IV)	Charge -Q at C and +Q at B	(S)	Electric potential at A is -ve
		(T)	Electric potential at C is +ve
		(U)	Electric potential at C is -ve

- (A) I $\rightarrow$ P,R,T; II $\rightarrow$ R,T; III $\rightarrow$ Q,S,U; IV $\rightarrow$ P,R,U
 (B) I $\rightarrow$ P,R,T; II $\rightarrow$ T; III $\rightarrow$ Q,S,U; IV $\rightarrow$ Q,R,U
 (C) I $\rightarrow$ P,S,T; II $\rightarrow$ S,T; III $\rightarrow$ Q,R,U; IV $\rightarrow$ P,S,U
 (D) I $\rightarrow$ P,R,S; II $\rightarrow$ T; III $\rightarrow$ Q,S,U; IV $\rightarrow$ Q,R,U

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

There is a spherical cavity of radius $R/3$ in a spherical solid conductor of radius R . A is centre of solid conductor, B is centre of cavity and C is a point outside at a distance $R/3$ from the surface. In List-I, there are certain charge configuration suggested and List-II discusses the electric field and potential. Now a charge $+Q$ is placed at B.



List-I		List-II	
(I)	Charge is moved slightly to left	(P)	Electric field at C changes.
(II)	Charge at B is made $+2Q$	(Q)	Electric potential at C increases.
(III)	Charge at B is made $-Q$	(R)	Electric potential at C decreases.
(IV)	An external uniform electric field is switched on in the direction C to A.	(S)	Electric field at A inside cavity changes.
		(T)	Electric potential at B due to charges induced on conductor changes
		(U)	Electric field at B due to charges induced on conductor changes.

- (A) I $\rightarrow$ P,T; II $\rightarrow$ P,Q,S,T; III $\rightarrow$ P,R,S,U; IV $\rightarrow$ P,R,S,T
 (B) I $\rightarrow$ Q,U; II $\rightarrow$ P,Q,S,U; III $\rightarrow$ P,R,T,U; IV $\rightarrow$ P,R,S,T,U
 (C) I $\rightarrow$ S,T; II $\rightarrow$ P,Q,S,T,U; III $\rightarrow$ P,R,S,T,U; IV $\rightarrow$ P,Q,T,U
 (D) I $\rightarrow$ S,U; II $\rightarrow$ P,Q,S,T; III $\rightarrow$ P,R,S,T; IV $\rightarrow$ P,R,T,U

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A planet is rotating about its axis. The radius of planet is R and its mass is m . It has a geostationary satellite in equatorial plane at a height h from surface.

List-I		List-II	
(I)	If the planet suddenly contracts without change in its mass	(P)	The energy of satellite decreases.
(II)	If the planet suddenly stop rotating.	(Q)	The energy of satellite increases.
(III)	If the geostationary satellite is pushed in direction of motion to suddenly double the velocity.	(R)	The energy of satellite remains same.
(IV)	If the geostationary satellite suddenly detaches half of its mass without any relative velocity.	(S)	The satellite covers larger area on the planet.
		(T)	The satellite does not remain geostationary.
		(U)	The satellite should have now been kept at a farther orbit from earth to be geostationary.

(A) $I \rightarrow R, S, T; II \rightarrow R, S, T; III \rightarrow Q, S, T; IV \rightarrow P$

(B) $I \rightarrow R, S, T; II \rightarrow R, S, T; III \rightarrow Q, S, T; IV \rightarrow Q$

(C) $I \rightarrow R, T, U; II \rightarrow R, S, T; III \rightarrow P, R, S, T; IV \rightarrow R$

(D) $I \rightarrow R, S, T; II \rightarrow R, S; III \rightarrow Q, S, T; IV \rightarrow P$

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A planet is rotating about its axis. The radius of planet is R and its mass is m . It has a geostationary satellite in equatorial plane at a height h from surface.

List-I		List-II	
(I)	Angular velocity of satellite.	(P)	Proportional to $\sqrt{R+h}$
(II)	Energy of satellite	(Q)	Proportional to $\frac{1}{\sqrt{R+h}}$
(III)	Angular momentum of satellite about centre of planet	(R)	Depends on mass of satellite.
(IV)	Speed of satellite	(S)	Depends on mass of planet.
		(T)	Proportional to $\sqrt{G}$
		(U)	Depends on angular velocity of planet.

- (A) $I \rightarrow S, T, U; II \rightarrow R, S, U; III \rightarrow P, R, S, T, U; IV \rightarrow Q, S, T, U$
 (B) $I \rightarrow S, T, U; II \rightarrow R, S; III \rightarrow P, R, S, T; IV \rightarrow Q, S, T$
 (C) $I \rightarrow R, T, U; II \rightarrow R, S; III \rightarrow P, R, S, T; IV \rightarrow Q, T, U$
 (D) $I \rightarrow P, S, T; II \rightarrow R, S, U; III \rightarrow P, R, S, T, U; IV \rightarrow Q, R, S, T$

Space for Rough Work

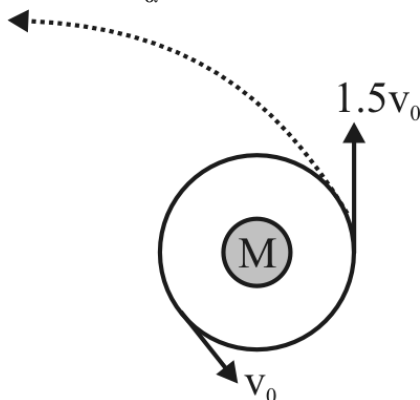
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

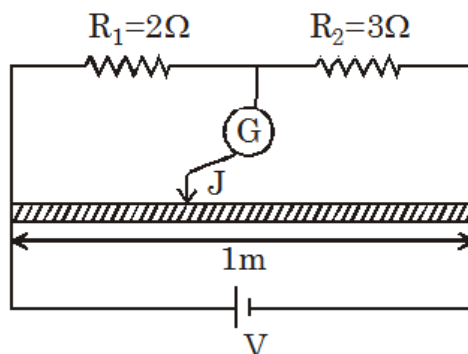
Zero Marks : 0 In all other cases.

1. An Earth satellite revolves on a circular orbit at a linear speed v_0 . A missile device increases the absolute value of this speed to $1.5v_0$, and the satellite goes into an unknown path as shown (Fig). With what speed will the satellite leave the gravitational field of earth? The resistance of the cosmic dust is not taken into account. Mass of earth is M . If your answer is $\frac{v_0}{\alpha}$, fill α in OMR sheet.

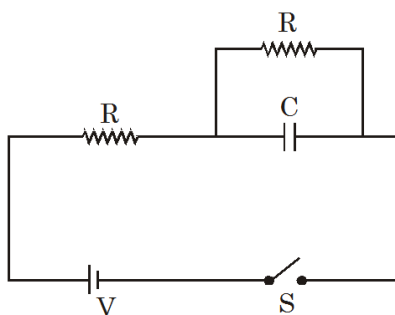


Space for Rough Work

2. In a meter bridge experiment, initially the jockey is at null point. Now resistance R_1 & R_2 is interchanged. Find the value (in cm) of shift in the position of jockey.



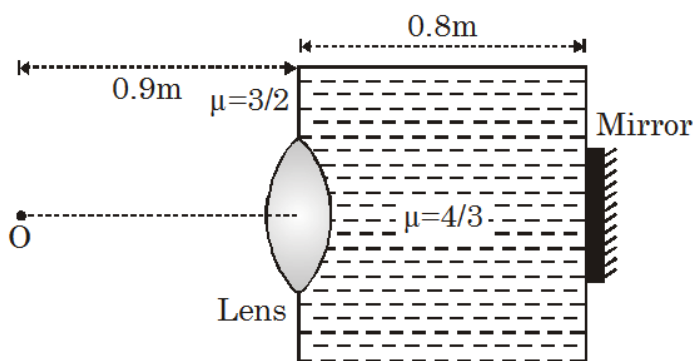
3. Write the variation in charge (q) on the capacitor plates as a function of time after the switch (S) is closed at $t = 0$. If q is given by $q = \frac{CV}{2} \left[1 - e^{-\frac{2t}{\alpha RC}} \right]$, then find the value of α .



4. A spherical shell is kept in an atmosphere at temperature T_0 . The wavelength corresponding to maximum intensity of radiation for the shell is λ_0 . A point source of constant power is switched on inside the shell. The power radiated by the source is $P = 0.4 \sigma S e T_0^4$ where S , e and σ are outer surface area of the shell, emissivity of the outer surface of the shell and Stefan's constant respectively. Calculate the new wavelength (λ) corresponding to the maximum intensity of radiation from the shell. Assume that change in temperature (ΔT) of the shell is small compared to the ambient temperature T_0 . If value of λ is given by $\left(\frac{N}{11} \right) \lambda_0$, then fill the value of N .

Space for Rough Work

5. A calorimeter of water equivalent 10 gm contains a liquid of mass 50 gm at 40°C . When 'm' gram of ice at -10°C is put into the calorimeter and the mixture is allowed to attain equilibrium, the final temperature was found to be 20°C . It is known that specific heat capacity of the liquid changes with temperature as $S = \left(1 + \frac{\theta}{500}\right) \text{ Cal g}^{-1} ^{\circ}\text{C}^{-1}$ where θ is temperature in $^{\circ}\text{C}$. The specific heat capacity of ice, water and calorimeter remains constant and values are $S_{\text{ice}} = 0.5 \text{ Cal g}^{-1} ^{\circ}\text{C}^{-1}$, $S_{\text{water}} = 1.0 \text{ Cal g}^{-1} ^{\circ}\text{C}^{-1}$ and latent heat of fusion of ice is $L_f = 80 \text{ Cal g}^{-1}$. Assume no heat loss to the surrounding and calculate the value of 'm' in gram.
6. A thin equiconvex lens of glass of refractive index $\mu = \frac{3}{2}$ and of focal length 0.3 m in air is sealed into an opening at one end of tank filled with water $\left(\mu = \frac{4}{3}\right)$. On the opposite side of the lens, a mirror is placed inside the tank on the tank wall perpendicular to the lens axis as shown in figure. The separation between the lens and the mirror is 0.8 m. A small object is placed outside the tank in front of the lens at a distance of 0.9 m from the lens along its axis. Find the position (relative to the lens) of final image formed by the system (in cm).



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. For an adiabatic irreversible process involving an ideal gas against constant external pressure correct option(s) is/are :

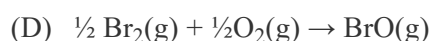
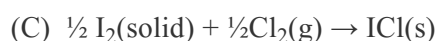
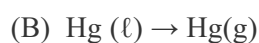
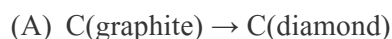
(A) $\Delta U = nC_{v,m}\Delta T$

(B) $W = nC_{v,m}\Delta T$

(C) $W = -P_{\text{ext.}}(V_2 - V_1)$

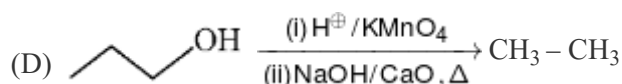
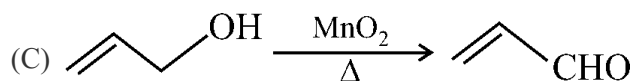
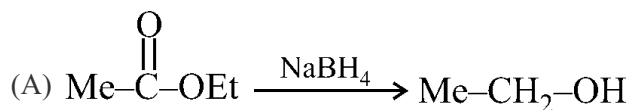
(D) $TV^{\gamma-1} = \text{constant}$

2. Which of the following reaction(s) represent standard enthalpy of formation of the product formed ?

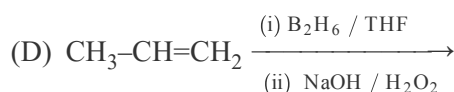
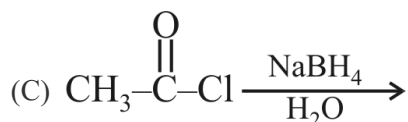


Space for Rough Work

3. Reaction correctly matched with their major product is/are :



4. Which of the following give 1° alcohol as a major product :



5. Identify correct statement (s) :

- (A) Magnetite (Fe_3O_4) is a ferrimagnetic substance
 (B) Transition metal oxides like TiO & CrO_2 show electrical conductance like metals
 (C) Solid oxide of composition $\text{Fe}_{0.95}\text{O}$ is a semi-conductor
 (D) Solid MnO can behave as an antiferromagnetic substance.

Space for Rough Work

6. In tricyclic meta phosphoric acid :

- (A) Number of P-O-P linkage = 6
- (B) Number of P-O-P linkage = 3
- (C) Number of P-P linkage = 3
- (D) Number of P-P linkage = 0

7. The IUPAC name of the complex $[(\text{NH}_3)_4\text{Co} \begin{array}{c} \text{NH}_2 \\ \diagup \quad \diagdown \\ \text{OH} \end{array} \text{Co}(\text{NH}_3)_4]^{4+}$ is :-

- (A) μ -amido- μ -hydroxidobis-[tetraamminecobalt(III)] ion.
- (B) Tetraamminecobalt(III)- μ -amido- μ -hydroxidotetraammine cobalt(III) ion
- (C) Tetraamminecobalt(II)- μ -amido- μ -hydroxidotetraammine cobalt(II) ion
- (D) μ -amido- μ -hydroxidobis-[tetraamminecobalt(II)] ion

8. Which of the following is/are diamagnetic and HOMO is gerade?

- (A) N_2
- (B) O_2
- (C) O_2^-
- (D) C_2

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I		List-II	
(I)	$\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_5\text{Cl}$	(P)	$\Delta H_{\text{mix}} = 0$
(II)	$\text{CH}_3\text{COCH}_3 + \text{C}_2\text{H}_5\text{OH}$	(Q)	$\Delta H_{\text{mix}} > 0$
(III)	$\text{CH}_3\text{COCH}_3 + \text{CHCl}_3$	(R)	$\Delta H_{\text{mix}} < 0$
(IV)	$\text{H}_2\text{O} + \text{C}_2\text{H}_5\text{OH}$	(S)	can form maximum boiling azeotrope
		(T)	can form minimum boiling azeotrope
		(U)	$\Delta V_{\text{mix}} > 0$

Which of the following is a **CORRECT** match ?

- (A) IV $\rightarrow$ R, S, U
 (B) II $\rightarrow$ Q, U
 (C) III $\rightarrow$ Q, T, U
 (D) I $\rightarrow$ R, S, U

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I		List-II	
(I)	$\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_5\text{Cl}$	(P)	$\Delta H_{\text{mix}} = 0$
(II)	$\text{CH}_3\text{COCH}_3 + \text{C}_2\text{H}_5\text{OH}$	(Q)	$\Delta H_{\text{mix}} > 0$
(III)	$\text{CH}_3\text{COCH}_3 + \text{CHCl}_3$	(R)	$\Delta H_{\text{mix}} < 0$
(IV)	$\text{H}_2\text{O} + \text{C}_2\text{H}_5\text{OH}$	(S)	can form maximum boiling azeotrope
		(T)	can form minimum boiling azeotrope
		(U)	$\Delta V_{\text{mix}} > 0$

Which of the following is an **INCORRECT** match ?

(A) II $\rightarrow$ Q, U

(B) I $\rightarrow$ P, U

(C) III $\rightarrow$ R, S

(D) IV $\rightarrow$ Q, T

Space for Rough Work

11. Match the list :

List-I		List-II	
(I)	$[\text{MnO}_4]^{-1}$	(P)	sp^3
(II)	$[\text{Ni}(\text{dmg})_2]$	(Q)	dsp^2
(III)	$[\text{CrO}_4]^{2-}$	(R)	d^3s
(IV)	$[\text{Zn}(\text{gly})_2]$	(S)	Paramagnetic
		(T)	Diamagnetic
		(U)	Coloured complex

Which of the following options has the correct combination considering List-I and List-II.

- (A) $\text{I} \rightarrow \text{Q}, \text{T}, \text{U}$ (B) $\text{II} \rightarrow \text{P}, \text{T}, \text{U}$ (C) $\text{I} \rightarrow \text{R}, \text{S}, \text{U}$ (D) $\text{II} \rightarrow \text{Q}, \text{T}, \text{U}$

12. Match the list :

List-I		List-II	
(I)	$[\text{MnO}_4]^{-1}$	(P)	sp^3
(II)	$[\text{Ni}(\text{dmg})_2]$	(Q)	dsp^2
(III)	$[\text{CrO}_4]^{2-}$	(R)	d^3s
(IV)	$[\text{Zn}(\text{gly})_2]$	(S)	Paramagnetic
		(T)	Diamagnetic
		(U)	Coloured complex

Which of the following options has the correct combination considering List-I and List-II.

- (A) $\text{III} \rightarrow \text{R}, \text{S}, \text{U}$ (B) $\text{IV} \rightarrow \text{Q}, \text{T}, \text{U}$ (C) $\text{III} \rightarrow \text{R}, \text{T}, \text{U}$ (D) $\text{IV} \rightarrow \text{P}, \text{S}, \text{U}$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

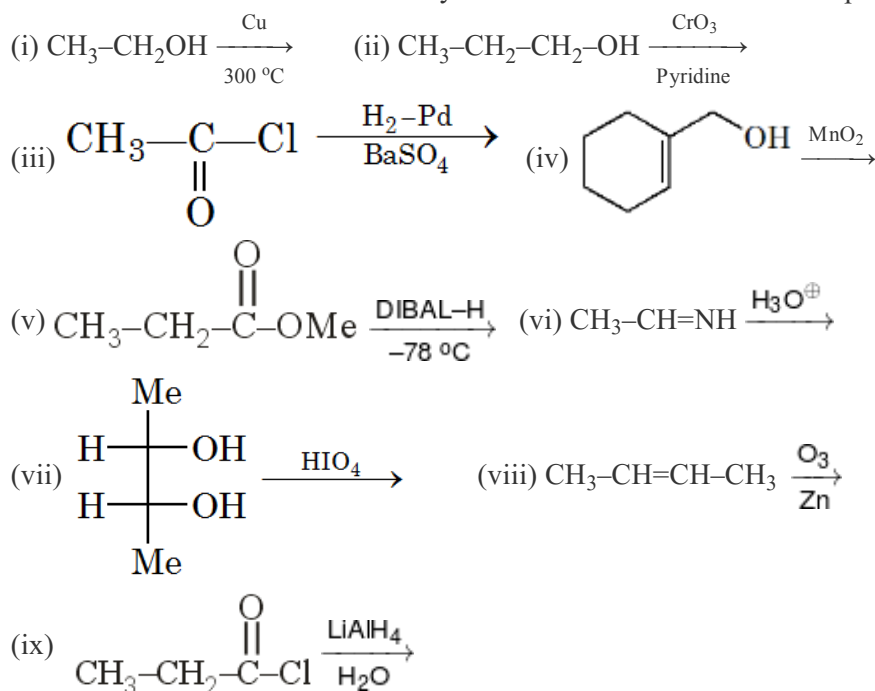
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

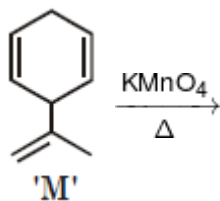
1. If rate of reaction becomes double when temperature is increased from 27°C to 37°C. Then find activation energy of reaction in kcal/mole.
($R = 2 \text{ cal/mol-K}$) ($\ln 2 = 0.7$)

2. Number of reactions in which aldehyde can be obtained as one of the product.

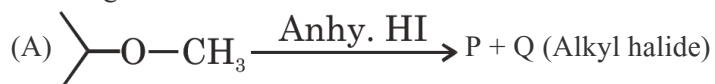


Space for Rough Work

3. Total number of moles of CO_2 evolved in following reaction per mole of compound 'M' is -



4. For the given reactions :



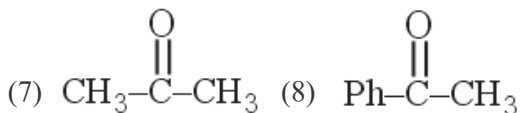
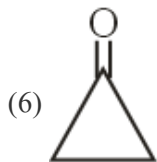
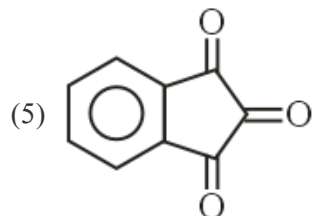
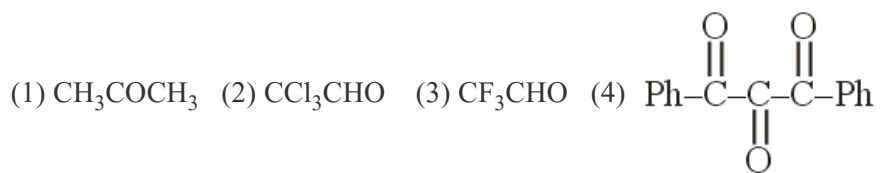
If the molar mass of P = a

and the molar mass of X = b

The value of $(a-b) / 2 = ?$

Space for Rough Work

5. How many of the following will form stable hydrate :



6. Total number of stereo isomer show by complex $[\text{Rh}(\text{en})\text{ClBrI}(\text{H}_2\text{O})]$:

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Two circles of radii a and 1 touch each other externally and are inscribed in the region bounded by $y = -\sqrt{4-x^2}$ and x -axis, then
- (A) $100(a+1) = 125$
- (B) Number of such pairs of circles is 2
- (C) $a = 1/2$
- (D) Points of contact of circle with radius a and $y = -\sqrt{4-x^2}$ are $\left(\frac{\pm 4\sqrt{2}}{3}, -\frac{2}{3}\right)$

Space for Rough Work

2. Which of the following statement(s) is/are correct ?

- (A) The point where the function changes its monotonicity and concavity is always the point of non-differentiability.
- (B) If the derivative of the function at $x = a$ is zero then $f(x)$ can be increasing or decreasing at that point.
- (C) If $f(x)$ is a twice differentiable function on $\mathbb{R}$ and has a relative maximum at $x = c$, then $f''(c)$ is negative.
- (D) If $f(x)$ is a differentiable function on $\mathbb{R}$ such that $f'(\alpha) = 0$, then $f(x)$ has an extremum at $x = \alpha$ ($\alpha \in \mathbb{R}$).

3. Let $f(x) = \begin{cases} x^3 + x^2 - 10x & ; -1 \leq x < 0 \\ \sin x & ; 0 \leq x < \frac{\pi}{2} \\ 1 + \cos x & ; \frac{\pi}{2} \leq x \leq \pi \end{cases}$ then $f(x)$ has

- (A) local maximum at $x = \frac{\pi}{2}$ (B) local minimum at $x = \frac{\pi}{2}$
(C) absolute maximum at $x = 0$ (D) absolute maximum at $x = -1$

Space for Rough Work

4. Let f be a differentiable function on $\mathbb{R}$ satisfying $f'(t) = e^t(\cos^2 t - \sin 2t)$ and $f(0) = 1$, then which of the following is/are correct ?
- (A) f is bounded in $(-\infty, 0)$.
- (B) Number of solution satisfying the equation $f(t) - e^t = 0$ in $[0, 2\pi]$ is 3.
- (C) The value of $\lim_{t \rightarrow 0} (f(t))^{\frac{1}{t}} = 1$
- (D) f is neither odd nor even.
5. Consider a function $f(x) = \int_{\pi}^{x^2} \frac{\sin x}{1 + \cos^2 \sqrt{\theta}} d\theta$ then which of the following is/are **CORRECT** ?
- (A) $f(x)$ has a local minima at $x = 2\pi$
- (B) $f(x)$ has a local maxima at $x = \pi$
- (C) $f(x)$ has a local minima at $x = 0$
- (D) $f(x)$ has a local maxima at $x = -\pi$
6. Consider the function g defined by $g(x) = \begin{cases} x^2 \sin \frac{\pi}{x} + (x-1)^2 \sin\left(\frac{\pi}{x-1}\right) & ; \quad x \neq 0, 1 \\ 0 & ; \quad \text{if } x = 0, 1 \end{cases}$ then which of the following statement(s) is/are correct ?
- (A) $g(x)$ is differentiable $\forall x \in \mathbb{R}$
- (B) $g'(x)$ is discontinuous at $x = 0$ but continuous at $x = 1$
- (C) $g'(x)$ is discontinuous at both $x = 0$ and $x = 1$
- (D) Rolle's theorem is applicable for $g(x)$ in $[0, 1]$

Space for Rough Work

7. Let $f(x) = \begin{cases} (x-2)^2 + 3 & x \geq 2 \\ k + x^2 & x < 2 \end{cases}$, then which of the following statements is/are true ?

- (A) If $f(x)$ is continuous at $x = 2$, then $k = -1$
 (B) There exists at least one value of k for which $f(x)$ is derivable $\forall x \in \mathbb{R}$
 (C) If $f(x)$ has a local minima at $x = 2$, then $k > -1$
 (D) If $f(x)$ is continuous at $x = 2$ then $k = 1$

8. Which of the following statement(s) is/are correct ?

- (A) If f is differentiable at $x = 1$, then $\lim_{x \rightarrow 1} \frac{f(x^3) - f(1)}{x - 1} = 3f'(1)$
 (B) If f is differentiable at $x = 0$, then $\lim_{x \rightarrow 0} \frac{f(ax) - f(bx)}{cx} = \frac{a-b}{c} f'(0)$
 (C) If f is differentiable at $x = x_0$, $\lim_{x \rightarrow x_0} \frac{xf(x_0) - x_0f(x)}{x - x_0} = f(x_0) - x_0f'(x_0)$
 (D) $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable and $f(x) \leq f(0) \forall x \in [-1, 1]$ then $f'(0) = 0$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
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 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

List-I		List-II	
(I)	$ax + by = 1$ is tangent to curve $x + y = xy$, where it cuts x-axis, then value of $a + b$ is	(P)	-7
(II)	Let $f(x)$ is continuous and differentiable over $[2, 9]$ and $f'(x) \in [-4, 3]$ $\forall x \in (2, 9)$, then sum of maximum and minimum value of $f(9) - f(2)$ is	(Q)	0
(III)	Let (α, β) is point of inflection for the curve $y = \frac{x^4 - 40}{12} + 16\ln\left(\frac{x}{2}\right)$, then value of $\alpha + \beta$ is	(R)	2
(IV)	Let $f(x) = \int_1^x (2t^2 + 3t - \lambda \log t) dt$ and $f'(x)$ is monotonic decreasing in interval $(0, 1]$, then least value of λ is	(S)	7
		(T)	-2
		(U)	1

Which of the following is the only correct combination ?

- (A) $I \rightarrow T$ (B) $I \rightarrow U$ (C) $II \rightarrow P$ (D) $II \rightarrow S$

Space for Rough Work

10.

List-I		List-II	
(I)	$ax + by = 1$ is tangent to curve $x + y = xy$, where it cuts x-axis, then value of $a + b$ is	(P)	-7
(II)	Let $f(x)$ is continuous and differentiable over $[2, 9]$ and $f'(x) \in [-4, 3]$ $\forall x \in (2, 9)$, then sum of maximum and minimum value of $f(9) - f(2)$ is	(Q)	0
(III)	Let (α, β) is point of inflection for the curve $y = \frac{x^4 - 40}{12} + 16 \ln\left(\frac{x}{2}\right)$, then value of $\alpha + \beta$ is	(R)	2
(IV)	Let $f(x) = \int_1^x (2t^2 + 3t - \lambda \log t) dt$ and $f'(x)$ is monotonic decreasing in interval $(0, 1]$, then least value of λ is	(S)	7
		(T)	-2
		(U)	1

Which of the following is the only correct combination ?

(A) III $\rightarrow$ R

(B) III $\rightarrow$ U

(C) IV $\rightarrow$ Q

(D) IV $\rightarrow$ S

Space for Rough Work

11. Let $f(x) = \begin{cases} x + a & ; x < 0 \\ \sqrt{x^2 - 2x + 1} & ; x \geq 0 \end{cases}$
 and $g(x) = \begin{cases} x + 1 & ; x < 0 \\ x^2 + 2(b - x) + 1 & ; x \geq 0 \end{cases}$,

a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all real x

List-I		List-II	
(I)	The value of b can be	(P)	0
(II)	The value of a can be	(Q)	1/2
(III)	Value of $\frac{d(g(f(x)))}{dx}$ at x = 4 is	(R)	1
(IV)	If f(x) is non differentiable at exactly k points, then the value of k is	(S)	2
		(T)	3
		(U)	4

Which of the following options is the only **CORRECT** combination ?

- (A) $I \rightarrow R$ (B) $I \rightarrow Q$ (C) $II \rightarrow Q$ (D) $II \rightarrow S$

Space for Rough Work

12. Let $f(x) = \begin{cases} x + a & ; x < 0 \\ \sqrt{x^2 - 2x + 1} & ; x \geq 0 \end{cases}$
 and $g(x) = \begin{cases} x + 1 & ; x < 0 \\ x^2 + 2(b - x) + 1 & ; x \geq 0 \end{cases}$,

a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all real x

List-I		List-II	
(I)	The value of b can be	(P)	0
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(III)	Value of $\frac{d(g(f(x)))}{dx}$ at $x = 4$ is	(R)	1
(IV)	If $f(x)$ is non differentiable at exactly k points, then the value of k is	(S)	2
		(T)	3
		(U)	4

Which of the following options is the only **CORRECT** combination ?

(A) III $\rightarrow$ P

(B) III $\rightarrow$ T

(C) IV $\rightarrow$ S

(D) IV $\rightarrow$ U

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If the line $y = 2 - x$ is tangent to the circle S at the point P(1,1) and circle S is orthogonal to the circle $x^2 + y^2 + 2x + 2y - 2 = 0$, then the length of tangent drawn from the point (2,2) to circle S is
2. If the value of the integral $\int_{1/2}^2 \frac{\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)}{x} dx$ is equal to $\pi \ln \sqrt{k}$ then 'k' is
3. Let f be a differentiable function on R and satisfying $f(x) = -(x^2 - x + 1)e^x + \int_0^x e^{x-y} \cdot f'(y) dy$. If $f(1) + f'(1) + f''(1) = ke$, where $k \in \mathbb{N}$, then value of $\frac{k}{2}$ is _____

Space for Rough Work

4. Let $f(x)$ be a differentiable function satisfying $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \quad \forall x, y \in \mathbb{R}$ and $f(0) = 0$. If $\int_0^{2\pi} (f(x) - \sin x)^2 dx$ is minimized, then find the value of $f(-4\pi^2)$?
5. If Rolle's theorem is applicable to the function f defined by $f(x) = \begin{cases} ax^2 + b & ; |x| < 1 \\ 1 & ; |x| = 1 \\ \frac{c}{|x|} & ; |x| > 1 \end{cases}$ for $x \in [-2, 2]$, then find the value of $(b^2 + c^2 + a^2)$.
6. If $\int (x^6 + 7x^5 + 6x^4 + 5x^3 + 4x^2 + 3x + 1)e^x dx$ is equal to $\sum_{k=1}^{\alpha} \beta x^k \cdot e^x + C$ (where C is constant of integration) then $\left(\frac{\alpha}{4\beta}\right)$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work